1. First understand the project name

AEX-TFANET

Break it into three parts:

AEX → Adaptive Explainable
T → Trust
FANET → Flying Ad Hoc Network

So:

> **AEX-TFANET = Adaptive Explainable Trust Framework for Secure Flying Ad Hoc Networks**

What is FANET?

A FANET is basically a network of **UAVs/drones communicating with each other**.

Imagine 5 drones flying in an area:

```
     UAV 1
     /   \
  UAV 2 --- UAV 3
    \       /
     UAV 4 - UAV 5
```

They can exchange information with each other.

Unlike a normal network, the drones are **moving continuously**.

Therefore:

- connections keep changing
- some drones move out of range
- new connections are created
- routes can break
- malicious drones can enter the network

That's why security becomes difficult.

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2. Slide: Problem Identification

What is the actual problem?

Suppose we have this:

UAV A → UAV B → UAV C → Destination

We assume UAV B will properly forward the packet.

But what if UAV B is malicious?

It could do:

UAV A → UAV B → ❌
Packet dropped

The sender doesn't immediately know whether B is:

- malicious,
- unreliable,
- experiencing communication problems,
- or simply moving away.

This is the main problem.

What you should say:

> "FANETs consist of highly mobile UAVs that communicate with each other without necessarily depending on fixed infrastructure. Because the UAVs are continuously moving, the network topology changes very frequently. This makes routing difficult. At the same time, malicious or unreliable UAVs can participate in communication and perform attacks such as packet dropping, blackhole or Sybil attacks. Therefore, we need a mechanism that can continuously evaluate the trustworthiness of UAVs and use that information for secure communication."

Then introduce your solution:

> "Our proposed AEX-TFANET framework addresses this problem using adaptive trust management combined with secure routing and explainable AI."

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3. Slide: Problem and Key Challenges

There are basically **three categories of challenges**.

A. Network challenge

The drones are moving.

Imagine:

10 seconds:

A ---- B ---- C

20 seconds:

A B
 C

The relationships have changed.

Therefore:

- routes break
- links change
- neighbours change

Say:

> "The first challenge is the dynamic nature of FANETs. Since UAVs move continuously, the network topology changes very quickly and existing routes may become unavailable."

B. Security challenge

Now suppose one UAV is bad.

A → B → C
 ↑
malicious UAV

It can:

- drop packets
- send false information
- pretend to be multiple UAVs
- disturb routing

Say:

> "The second challenge is security. A malicious UAV can participate in the network and affect communication by dropping packets, providing false information or disrupting routing."

C. Intelligence challenge

Even if AI detects a malicious UAV, another question arises:

Why did AI classify it as malicious?

For example:

> Trust = 0.25

But why?

Maybe:

- packet forwarding = poor
- link stability = poor
- behaviour = suspicious

This is where **Explainable AI** comes in.

---

4. Slide: Objectives

Don't try to explain every bullet individually.

Your objective is basically:

Observe → Trust → Detect → Route → Explain

You can say:

> "Our first objective is to continuously monitor UAV behaviour and calculate a trust score. Based on this trust score, we identify reliable and potentially malicious nodes. We then use the trust information during route selection so that communication prefers more trustworthy UAVs. Finally, we incorporate Explainable AI so that the system can explain why a particular UAV received a certain trust score or why a particular route was selected."

That's enough.

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5. Slide: Literature Review

Your teacher may ask:

"What did previous researchers do?"

You should understand the progression.

Traditional research

People first focused on:

Routing

How do we find a path?

A → B → C → D

Then researchers started asking:

"Can we trust B and C?"

So came:

Trust-based routing

A → B → C → D
  ↑
 trust

Then:

"Can AI make this decision automatically?"

So:

AI-based security

Then:

"But can we understand AI's decision?"

So:

Explainable AI

Your project brings these ideas together.

Say:

> "From the literature, we found that trust-based approaches can help identify unreliable UAVs, while AI-based approaches can improve adaptive decision-making. However, AI decisions can sometimes be difficult to interpret. Therefore, our work focuses on combining adaptive trust management, secure routing and explainability into a unified framework."

---

6. Slide: Literature Comparison

This slide is actually very easy.

Think of it as evolution:

Traditional Routing
↓
Cryptographic Security
↓
Trust-Based Routing
↓

AI-Based Security
↓
Explainable AI
↓
OUR
AEX-TFANET

Why isn't traditional routing enough?

Because it asks:

> "Which path is available?"

Your system asks:

> **"Which path is available AND trustworthy?"**

That's an important difference.

---

7. Slide: Research Gap

This is probably one of the **most important slides**.

What is a research gap?

It means:

> **What existing research has not adequately solved yet?**

Your gap is:

Existing systems may provide:

- routing
- security
- trust
- AI

But your proposed system tries to combine:

Trust + Adaptation + Secure Routing + Explainability

For example:

Suppose:

UAV A
/ \
B C
\ /
D

Maybe B is highly trusted and C is suspicious.

AEX-TFANET should understand that and prefer:

A → B → D

instead of:

A → C → D

And it should be able to explain:

> "B was selected because it has higher trust based on packet forwarding behaviour and communication reliability."

That's the **explainability** part.

---

8. Slide: Proposed Methodology

This is the **heart of your project**.

You should understand this flow extremely well:

UAVs
↓
Monitor Behaviour
↓
Collect Data
↓
Calculate Trust
↓
Update Trust Adaptively
↓

Detect Malicious Nodes
↓
Select Secure Route
↓
Explain Decision

Let's understand each.

Step 1 — FANET Initialization

We create our UAV network.

Example:

UAV1
UAV2
UAV3
UAV4
UAV5

They communicate with each other.

Step 2 — Network Monitoring

We observe what the UAVs are doing.

For example:

> Is UAV2 forwarding packets?
>
> Is UAV3 behaving consistently?
>
> Is the communication link stable?

Step 3 — Trust Calculation

Now we calculate a **trust score**.

For example:

UAV	Trust
UAV1	0.92
UAV2	0.84
UAV3	0.31
UAV4	0.76

So UAV3 looks suspicious.

The exact mathematical formula isn't defined in the source material, so don't invent a specific equation during your presentation.

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9. What does "Adaptive Trust" mean?

This is a **very important question your teacher may ask**.

Suppose UAV3 initially has:

Trust = 0.8

Then it starts dropping packets.

The system observes this.

Its trust could gradually become:

0.8
↓
0.65
↓
0.45
↓
0.25

So the trust value **changes according to new behaviour**.

That's why it's called:

Adaptive Trust

It doesn't permanently say:

> "UAV3 is trustworthy."

Instead:

> "I'll keep evaluating UAV3 based on its latest behaviour."

---

10. What is Direct and Indirect Trust?

Your PPT mentions direct and indirect trust.

Direct trust

You personally observe a UAV.

Example:

> UAV A sends 100 packets through UAV B.

B successfully forwards 95.

A can use this behaviour to estimate B's trust.

Indirect trust

Suppose A doesn't have enough information about B.

Another UAV says:

> "I've interacted with B before. B has behaved reliably."

That recommendation can contribute to trust evaluation.

So:

Direct Trust
+
Indirect Trust

↓
Overall Trust

---

11. Secure Route Selection

Now we use the trust value.

Suppose there are two possible routes:

Route 1:
A → B → D

Route 2:
A → C → D

Trust:

B = 0.90
C = 0.30

The system should prefer:

A → B → D

because it is more trustworthy.

Important:

Don't say:

> "The system always chooses the highest trust node."

Because routing also depends on network conditions.

Say:

> "Trust becomes an important factor in selecting a reliable and secure route."

---

12. Malicious Node Detection

Suppose UAV C repeatedly:

- drops packets
- behaves abnormally
- provides inconsistent information

Its trust decreases.

Eventually:

Trust(C) < Threshold

Then the system can classify C as suspicious/untrusted.

Conceptually:

Normal
↓
Suspicious
↓
Low Trust
↓
Malicious / Untrusted

---

13. Explainable AI

This is probably the **most interesting part of your project**.

Imagine your system says:

UAV C = Malicious

A normal AI system might just give you the answer.

But XAI tries to explain:

Why?

For example:

UAV C Trust Score = 0.27

Reasons:
Packet forwarding ↓

Link stability ↓
Behaviour consistency ↓
Historical trust ↓

Therefore:

> "The UAV was classified as suspicious mainly because of poor packet forwarding and inconsistent behaviour."

That's what makes the system **explainable**.

---

14. SHAP and LIME

Your technology stack contains:

- SHAP
- LIME

Don't get scared by these names.

Their purpose is basically:

"Which features influenced the AI's decision?"

Suppose the AI predicts:

> UAV = suspicious

SHAP/LIME can help explain which inputs contributed to that prediction.

For example:

Packet Forwarding +++
Link Stability ++
Historical Behaviour +++
Mobility +

So the operator can understand the decision rather than blindly trusting the AI.

If teacher asks:

"What is SHAP?"

Say:

> "SHAP is an Explainable AI technique that helps identify how individual input features contribute to a model's prediction."

"What is LIME?"

> "LIME explains an individual prediction by approximating the model's behaviour locally around that particular instance."

That's enough for a B.Tech presentation.

---

15. Trust Classification

Your slide has four levels.

Highly Trusted

Trust ≈ High

Normal and reliable behaviour.

Trusted

Generally reliable.

Suspicious

Something unusual is happening.

Malicious / Untrusted

Repeated or strong evidence of abnormal behaviour.

Think:

HIGH TRUST
↓
TRUSTED
↓
SUSPICIOUS
↓

UNTRUSTED

---

16. System Architecture

This is basically your entire project in one slide.

You should explain it from **left/top to bottom**.

UAV Nodes

↓

They communicate with each other.

Network Monitoring

↓

We collect information about their behaviour.

Trust Data Collection

↓

We collect parameters relevant to trust.

Adaptive Trust Engine

↓

Calculate/update trust.

Malicious Node Detection

↓

Identify suspicious nodes.

Secure Route Selection

↓

Avoid unreliable paths and prefer trustworthy communication.

Explainable AI

↓

Explain why the decision was made.

Security Decision

↓

Take appropriate action.

---

17. Technology Stack

You need to know **why** you're using each technology.

NS-3

Used for:

> **Network simulation**

We can simulate UAVs and their communication.

Python

Used for:

> Trust management, AI/ML and analysis.

SHAP/LIME

Used for:

> Explainability.

NumPy/Pandas

Used for:

> Data processing and analysis.

Matplotlib

Used for:

> Graphs and performance visualization.

---

18. Work Plan

This slide is simply:

> **What have we done and what are we going to do?**

You have:

Completed

- Problem identification
- Literature review

Planned

- FANET design
- Trust model
- Adaptive engine
- Secure routing
- XAI module
- Simulation
- Testing
- Integration

If your actual project has progressed further by presentation day, **change the status accordingly**. Don't claim something is completed if it isn't.

---

19. Expected Outcome

The important thing here is:

You are proposing a framework.

Therefore don't say:

> "Our system achieved 95% detection accuracy."

unless you actually have experimental results.

Instead say:

> "We expect the framework to improve identification of unreliable UAVs, support trust-aware route selection and provide interpretable security decisions."

That's academically safer.

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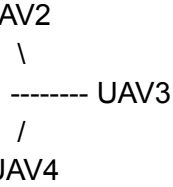
20. The entire project in ONE example

If your teacher says:

"Explain your project in simple words."

Give this example.

Imagine there are **10 drones** communicating with each other.



UAV1 wants to send a packet to UAV4.

There are multiple possible routes.

Our system observes the behaviour of UAV2 and UAV3.

Suppose:

UAV2 → Trust = 0.90
UAV3 → Trust = 0.35

UAV3 has been dropping packets.

The system therefore considers UAV2 more reliable and prefers a route involving UAV2.

At the same time, the system can explain:

> "UAV2 was preferred because of its high packet-forwarding reliability and stable communication behaviour."

If UAV3 continues behaving badly:

Trust:
0.60 → 0.45 → 0.35 → 0.20

the system can classify it as suspicious or untrusted and avoid using it.

THAT is essentially your entire project.

---

21. Your 1-minute project explanation

If you get:

> **"Tell me about your project."**

Say this:

> "Our project is AEX-TFANET, which stands for Adaptive Explainable Trust Framework for Secure Flying Ad Hoc Networks. FANETs consist of UAVs communicating with each other, but because UAVs are highly mobile, the network topology changes frequently and malicious or unreliable nodes can disrupt communication. Our proposed framework continuously monitors UAV behaviour and calculates a trust score based on factors such as packet forwarding, communication reliability and behavioural history. This trust is updated adaptively as the behaviour of the UAV changes. The trust information is then used during secure route selection so that unreliable nodes can be avoided. In addition, we incorporate Explainable AI techniques such as SHAP and LIME to explain why a UAV was considered trustworthy, suspicious or malicious. The overall goal is to make UAV communication more secure, adaptive and transparent."

That is the answer I would memorize.

---

22. Most likely questions from your teacher

Be prepared for these:

Q1. What is FANET?

> "FANET stands for Flying Ad Hoc Network. It is a network in which UAVs communicate with each other, often without depending completely on fixed infrastructure."

Q2. Why is FANET different from MANET?

> "The major difference is the mobility and operating environment. UAVs generally move much faster and in three-dimensional space, causing more frequent topology changes."

Q3. Why do we need trust?

> "Because a UAV participating in the network may not always behave reliably. Trust helps us distinguish reliable nodes from suspicious or potentially malicious nodes."

Q4. What is adaptive trust?

> "It means the trust value is continuously updated according to the latest observed behaviour of a UAV."

Q5. Why not simply use encryption?

> "Encryption protects the confidentiality and integrity of communication, but it does not necessarily tell us whether an authenticated UAV is behaving maliciously. Trust management addresses behavioural reliability."

This is a very good answer.

Q6. Why do you need Explainable AI?

> "Because an AI model may make a prediction without clearly showing why. XAI helps us understand which factors contributed to the security decision."

Q7. What is SHAP?

> "SHAP explains the contribution of individual features to a model's prediction."

Q8. What is LIME?

> "LIME explains an individual prediction by building a simpler local approximation around that particular prediction."

Q9. What attacks are you considering?

You can mention:

- Blackhole
- Packet dropping
- Sybil
- Denial-of-Service
- False information

But **don't claim you have implemented all of them** unless you actually have.

Q10. What is the novelty?

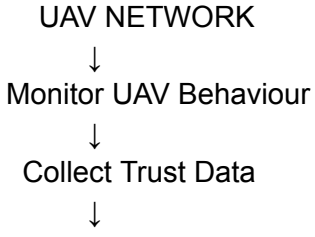
This is an important one.

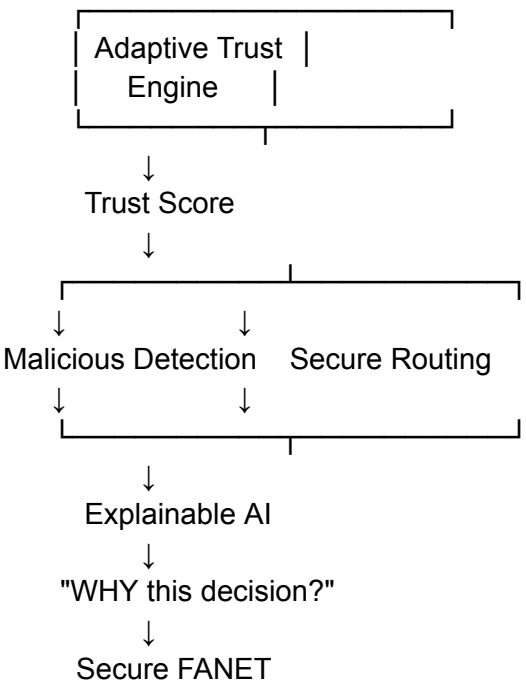
Say:

> "The proposed novelty is the combination of adaptive trust management, trust-aware secure routing and explainability within a unified FANET security framework. Instead of only detecting an unreliable node, the framework also adapts trust according to behaviour and provides an explanation for the resulting security decision."

The one diagram you should remember

If you remember only one thing before presenting, remember this:





And the simplest possible definition of your project is:

> **"We want UAVs to trust the right neighbours, avoid unreliable ones, choose safer routes, and explain why those decisions were made."**

If you understand that sentence, you understand the core of **AEX-TFANET**.